

1. [20 pts] [Basic Concept of “Electromagnetics”] Please succinctly answer for the following questions:

(a) [4 pts] “What is “Electromagnetics”?”

The study of the effects of electric charges at rest and in motion

(b) [4 pts] What are the four fundamental SI unit in the electromagnetics?

Meter, Kilogram, Second, Ampere

(c) [6 pts] What are the four fundamental field quantities in the electromagnetics model? and write their units

E: Electric Field Intensity V/m

D: Electric Flux Density (per unit area) C/m²

H: Magnetic Field Intensity A/m

B: Magnetic Flux Density Tesla

(d) [6 pts] What are the three-universal constants in the electromagnetic model? and find the permittivity in free space, ϵ_0 , of material from the physical relationship between c and magnetic permeability in free space, μ_0

$$c = 3 * 10^8 \text{ m/sec}$$

$$\mu_0 = \frac{B}{H} = 4\pi * 10^{-7} \text{ H/m}$$

$$\epsilon_0 = \frac{1}{36\pi} * 10^{-9} \text{ F/m}$$

$$c = \frac{1}{\sqrt{\epsilon_0 * \mu_0}} \rightarrow \epsilon_0 = \frac{1}{\mu * c^2} \rightarrow \frac{1}{4\pi * 10^{-7} * (3 * 10^8)^2} = \frac{1}{36 * 10^9 \pi} = \frac{1}{36\pi} * 10^{-9}$$

2. [20 pts] [Understanding of Vector Presentation in terms of Orthogonal Rectangular Components] In rectangular coordinates, given points $M(-1, 2, 1)$, $N(3, -3, 0)$, and $P(-2, -3, -4)$. Find:

(a) [5 pts] $\overrightarrow{A_{MN}}$, Magnitude of $\overrightarrow{A_{MN}}$, and Direction of Unit Vector, $\widehat{a_{MN}}$

$$\overrightarrow{A_{MN}} = \langle 3 - (-1), -3 - 2, 0 - 1 \rangle = \langle 4, -5, -1 \rangle$$

$$|\overrightarrow{A_{MN}}| = \sqrt{4^2 + (-5)^2 + (-1)^2} = \sqrt{42} \cong 6.48$$

$$\mathbb{u} = \frac{\mathbb{v}}{|\mathbb{v}|} = \widehat{a_{MN}} = \frac{\langle 4, -5, -1 \rangle}{\sqrt{42}} = \left\langle \frac{4}{\sqrt{42}}, \frac{-5}{\sqrt{42}}, \frac{-1}{\sqrt{42}} \right\rangle \cong \langle 0.617, -0.772, -0.154 \rangle$$

(b) [5 pts] $\overrightarrow{A_{MN}} + \overrightarrow{A_{MP}}$

$$\overrightarrow{A_{MN}} = \langle 3 - (-1), -3 - 2, 0 - 1 \rangle = \langle 4, -5, -1 \rangle$$

$$\overrightarrow{A_{MP}} = \langle -2 - (-1), -3 - 2, -4 - 1 \rangle = \langle -1, -5, -5 \rangle$$

$$\overrightarrow{A_{MN}} + \overrightarrow{A_{MP}} = \langle -1, -5, -5 \rangle + \langle 4, -5, -1 \rangle = \langle 3, -10, -6 \rangle$$

(c) [5 pts] $|\overrightarrow{A_M}|$, and $|\overrightarrow{A_N}|$

$$\overrightarrow{A_M} = \langle -1, 2, 1 \rangle$$

$$\overrightarrow{A_N} = \langle 3, -3, 0 \rangle$$

$$|\overrightarrow{A_M}| = \sqrt{(-1)^2 + 2^2 + 1^2} = \sqrt{8} \cong 2.8284$$

$$|\overrightarrow{A_N}| = \sqrt{3^2 + (-3)^2 + 0^2} = \sqrt{18} \cong 4.2426$$

(d) [5 pts] $|5\overrightarrow{A_P} - 2\overrightarrow{A_N}| * (\overrightarrow{A_{MP}} + \overrightarrow{A_{PN}})$

$$\overrightarrow{A_P} = \langle -2, -3, -4 \rangle$$

$$\overrightarrow{A_N} = \langle 3, -3, 0 \rangle$$

$$5\overrightarrow{A_P} - 2\overrightarrow{A_N} = 5 * \langle -2, -3, -4 \rangle - 2 * \langle 3, -3, 0 \rangle = \langle (-10 - 6), (-15 - (-6)), (-20 - 0) \rangle \\ = \langle -16, -9, -20 \rangle$$

$$|5\overrightarrow{A_P} - 2\overrightarrow{A_N}| = \sqrt{(-16)^2 + (-9)^2 + (-20)^2} = \sqrt{737} \cong 27.1477$$

$$\overrightarrow{A_{MP}} = \langle -2 - (-1), -3 - 2, -4 - 1 \rangle = \langle -1, -5, -5 \rangle$$

$$\overrightarrow{A_{PN}} = \langle 3 - (-2), -3 - (-3), 0 - (-4) \rangle = \langle 5, 0, 4 \rangle$$

$$\overrightarrow{A_{MP}} + \overrightarrow{A_{PN}} = \langle -1, -5, -5 \rangle + \langle 5, 0, 4 \rangle = \langle 4, -5, -1 \rangle$$

$$|5\overrightarrow{A_P} - 2\overrightarrow{A_N}| * (\overrightarrow{A_{MP}} + \overrightarrow{A_{PN}}) = \sqrt{737} * \langle 4, -5, -1 \rangle \cong \langle 108.59097, -135.73872, -27.14774 \rangle$$

3. [30 pts] [Understanding of Vector Field and Dot and Cross Products] In Cartesian coordinates, vector $\vec{A}$ is directed from the origin to point $P_1 (2, 3, 3)$, and vector $\vec{B}$ is directed from P_1 to point $P_2 (1, -2, 2)$. Find:

(a) [3 pts] vector $\vec{A}$, its magnitude A , and unit vector $\hat{a}$,

$$\vec{A} = \langle 2, 3, 3 \rangle$$

$$|\vec{A}| = \sqrt{(2)^2 + (3)^2 + (3)^2} = \sqrt{22} \cong 4.69$$

$$\hat{a} = \frac{\vec{A}}{|\vec{A}|} = \frac{\langle 2, 3, 3 \rangle}{\sqrt{22}} = \langle \frac{2}{\sqrt{22}}, \frac{3}{\sqrt{22}}, \frac{3}{\sqrt{22}} \rangle \cong \langle 0.4264, 0.6396, 0.6396 \rangle$$

(b) [4 pts] the angle that $\vec{A}$ makes with the y-axis,

$$\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{Y}}{|\vec{A}| * |\vec{Y}|} \right)$$

Create a vector parallel to the y-axis

$$\vec{Y} = \langle 0, 1, 0 \rangle$$

$$|\vec{A}| = \sqrt{(2)^2 + (3)^2 + (3)^2} = \sqrt{22} \cong 4.69$$

$$|\vec{Y}| = 1$$

$$\vec{A} \cdot \vec{Y} = 0 * 2 + 1 * 3 + 0 * 3 = 3$$

$$\theta = \cos^{-1} \left(\frac{3}{1 * \sqrt{22}} \right) = 50.2378^\circ$$

(c) [3 pts] vector $\vec{B}$

$$\vec{B} = \langle (1 - 2), (-2 - 3), (2 - 3) \rangle = \langle -1, -5, -1 \rangle$$

(d) [4 pts] the angle between $\vec{A}$ and $\vec{B}$

$$\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}| * |\vec{B}|} \right)$$

Create a vector parallel to the y-axis

$$\vec{B} = \langle -1, -5, -1 \rangle$$

$$|\vec{A}| = \sqrt{(2)^2 + (3)^2 + (3)^2} = \sqrt{22} \cong 4.69$$

$$|\vec{B}| = \sqrt{(-1)^2 + (-5)^2 + (-1)^2} = \sqrt{27} \cong 5.196$$

$$\vec{A} \cdot \vec{B} = (-1) * 2 + (-5) * 3 + (-1) * 3 = -20$$

$$\theta = \cos^{-1}\left(\frac{-20}{\sqrt{27} * \sqrt{22}}\right) = 145.145898^\circ$$

(e) [4 pts] the perpendicular distance from the origin to vector $\vec{B}$

$$\theta = \cos^{-1}\left(\frac{\vec{V} \cdot \vec{B}}{|\vec{V}| * |\vec{B}|}\right) = 90^\circ \rightarrow \frac{\vec{V} \cdot \vec{B}}{|\vec{V}| * |\vec{B}|} = 0$$

$$\vec{B} = \langle -1, -5, -1 \rangle$$

The point where the perpendicular line will intersect vector $\vec{B}$ is modeled by this equation:

$$\vec{P}_x = P_1 + \vec{B} * x, \text{ where } x \text{ is the distance the point is from } P_1$$

$$\vec{P}_x = (2 - 1 * x), (3 - 5 * x), (3 - 1 * x)$$

The perpendicular Vector will be from P_x to the origin, so it is modeled by the following formula

$$\vec{V} = 0 - (2 - x), 0 - (3 - 5x), 0 - (3 - x) = \langle (x - 2), (5x - 3), (x - 3) \rangle$$

$$|\vec{V}| = \sqrt{(x - 2)^2 + (5x - 3)^2 + (x - 3)^2}$$

$$|\vec{B}| = \sqrt{(-1)^2 + (-5)^2 + (-1)^2} = \sqrt{27}$$

$$\vec{V} \cdot \vec{B} = (-1) * (x - 2) + (-5) * (5x - 3) + (-1) * (x - 3)$$

Solve for x:

$$\frac{(-1) * (x - 2) + (-5) * (5x - 3) + (-1) * (x - 3)}{\sqrt{(x - 2)^2 + (5x - 3)^2 + (x - 3)^2} + \sqrt{27}} = 0 \rightarrow$$

$$(-1) * (x - 2) + (-5) * (5x - 3) + (-1) * (x - 3) = 0 \rightarrow$$

$$(2 - x) + (15 - 25x) + (3 - x) = 0 \rightarrow 20 - 27x = 0 \rightarrow x = \frac{20}{27}$$

Input x back into Vector V to find the magnitude

$$\vec{V} = \langle (x - 2), (5x - 3), (x - 3) \rangle \rightarrow \left\langle \left(\frac{20}{27} - 2\right), \left(5 * \frac{20}{27} - 3\right), \left(\frac{20}{27} - 3\right) \right\rangle$$

$$|\vec{V}| = \sqrt{\left(\frac{20}{27} - 2\right)^2 + \left(5 * \frac{20}{27} - 3\right)^2 + \left(\frac{20}{27} - 3\right)^2} = \frac{\sqrt{582}}{9} \cong 2.6805$$

Book Solution:

$$|\vec{V}| = |\vec{A}| * \sin(180^\circ - \theta_{AB}) \rightarrow \sqrt{22} * \sin(180^\circ - 145.145898^\circ) \cong 2.68$$

(f) [6 pts] $\vec{A} \cdot \vec{B}$, and $\vec{B} \cdot \vec{A}$. Compare the calculation results are the same or not.

$$\vec{A} = \langle 2, 3, 3 \rangle$$

$$\vec{B} = \langle -1, -5, -1 \rangle$$

$$\vec{A} \cdot \vec{B} = (-1) * 2 + (-5) * 3 + (-1) * 3 = -20$$

$$\vec{B} \cdot \vec{A} = 2 * (-1) + 3 * (-5) + 3 * (-1) = -20$$

They are the same.

(g) [6 pts] $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$

$$\vec{A} = \langle 2, 3, 3 \rangle$$

$$\vec{B} = \langle -1, -5, -1 \rangle$$

$$\vec{A} \times \vec{B} = \langle (3 * (-1) - 3 * (-5)), (3 * (-1) - 2 * (-1)), (2 * (-5) - 3 * (-1)) \rangle \rightarrow \langle 12, -1, -7 \rangle$$

$$\vec{B} \times \vec{A} = \langle (-5 * 3 - (-1) * 3), (-1 * 2 - (-1) * 3), (-1 * 3 - (-5) * 2) \rangle \rightarrow \langle -12, 1, 7 \rangle$$

They are NOT the same

4. [10 pts] [Understanding of Basic Law of Cross Vector Products] Given $\vec{A} = \hat{x} - \hat{y} + \hat{z}$, $\vec{B} = \hat{y} + \hat{z}$, and $\vec{C} = -\hat{x}\hat{z} + \hat{z}\hat{3}$. Find $(\vec{A} \times \vec{B}) \times \vec{C}$ and compare with $\vec{A} \times (\vec{B} \times \vec{C})$.

$$\vec{A} = \langle 1, -1, 1 \rangle$$

$$\vec{B} = \langle 0, 1, 1 \rangle$$

$$\vec{C} = \langle -1, 0, 1 \rangle$$

$$\vec{A} \times \vec{B} = \langle ((-1) * 1 - 2 * 1), (2 * 0 - 1 * 1), (1 * 1 - 0 * (-1)) \rangle \rightarrow \langle -3, -1, 1 \rangle$$

$$(\vec{A} \times \vec{B}) \times \vec{C} = \langle ((-1) * 1 - 1 * 0), (1 * (-1) - (-3) * 1), ((-3) * 0 - (-1) * (-1)) \rangle \rightarrow \langle -1, 2, -1 \rangle$$

$$\vec{B} \times \vec{C} = \langle (1 * 1 - 1 * 0), (1 * (-1) - 0 * 1), (0 * 0 - 1 * (-1)) \rangle \rightarrow \langle 1, -1, 1 \rangle$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = \langle ((-1) * 1 - 2 * (-1)), (2 * 1 - 1 * 1), (1 * (-1) - (-1) * 1) \rangle \rightarrow \langle 1, 1, 0 \rangle$$

5. [20 pts] (HARD PROBLEM) [Understanding of Two Vector Positions Based on the Geometrical

Properties of Dot and Cross Products] Given $\vec{A} = \hat{a}_x 5 - \hat{a}_y 2 + \hat{a}_z$, find the expression of a unit vector $\vec{B}$ in the case under below situations

(a) $\vec{B} \parallel \vec{A}$

Because $\vec{B}$ is parallel to $\vec{A}$, we can say $\vec{B} = \vec{A} * x$

$$\vec{B} = x * \langle 5, -2, 1 \rangle = \langle 5x, -2x, x \rangle$$

$$|\vec{B}| = 1;$$

$$\sqrt{(5x)^2 + (-2x)^2 + x^2} = 1$$

$$x = \pm \frac{1}{\sqrt{30}}$$

$$\vec{B} = \left\langle \frac{5}{\sqrt{30}}, \frac{-2}{\sqrt{30}}, \frac{1}{\sqrt{30}} \right\rangle$$

(b) $\vec{B} \perp \vec{A}$, if $\vec{B}$ lies in the xy-plane

Because $\vec{B}$ lies in the xy-plane, we can create its vector as such: $\langle x, y, 0 \rangle$

Because $\vec{B}$ is perpendicular to $\vec{A}$, we know that $\vec{A} \cdot \vec{B} = 0$

We also know that $|\vec{B}| = 1$

$$\langle 5, -2, 1 \rangle \cdot \langle x, y, 0 \rangle = 0 \rightarrow 5x - 2y = 0$$

$$\sqrt{x^2 + y^2} = 1 \rightarrow x^2 + y^2 = 1$$

Solve system of equations:

$$5x - 2y = 0 \rightarrow 5x = 2y \rightarrow 25x^2 = 4y^2 \rightarrow 25x^2 - 4y^2 = 0$$

To simplify the system of equations, set $x^2 = a, y^2 = b$

$$\begin{aligned} (25a - 4b = 0) \\ (a + b = 1) \end{aligned} \rightarrow a = (1 - b) \rightarrow 25(1 - b) - 4b = 0 \rightarrow b = \frac{25}{29}, \text{ and } a = 1 - \frac{25}{29} = \frac{4}{29}$$

$$x^2 = \frac{4}{29}, \text{ thus } x = \sqrt{\frac{4}{29}}, y^2 = \frac{25}{29}, \text{ thus } y = \sqrt{\frac{25}{29}}$$

Now, recall that Vector $\vec{B}$ is equal to $\langle x, y, 0 \rangle$, and thus, substituting the previous results into the vector,

we find that the result is $\left\langle \sqrt{\frac{4}{29}}, \sqrt{\frac{25}{29}}, 0 \right\rangle$